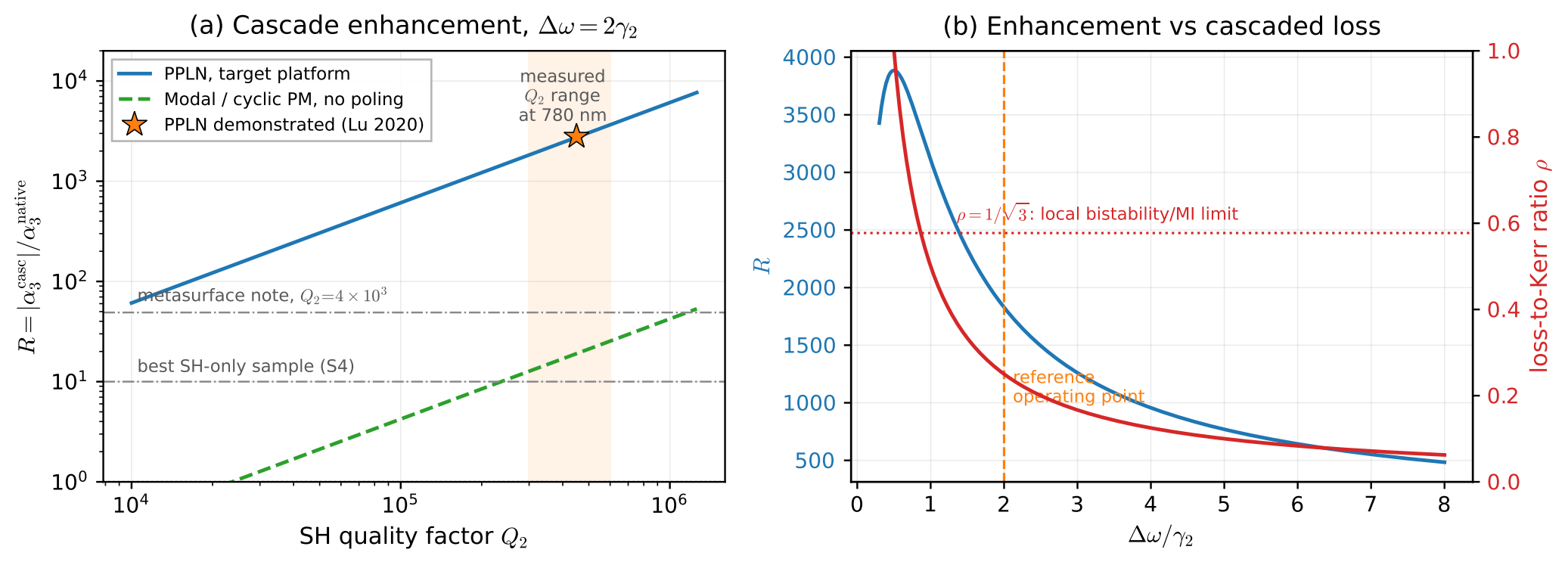
(a) Cascade enhancement, $\Delta\omega = 2\gamma_2$
PPLN, target platform
Modal / cyclic PM, no poling
PPLN demonstrated (Lu 2020)
measured
Q_2 range
at 780 nm
metasurface note, $Q_2 = 4 \times 10^3$
best SH-only sample (S4)
$R = |\alpha_3^{\mathrm{casc}}|/\alpha_3^{\mathrm{native}}$
SH quality factor Q_2
(b) Enhancement vs cascaded loss
R
loss-to-Kerr ratio ρ
$\rho = 1/\sqrt{3}$: local bistability/MI limit
reference
operating point
$\Delta\omega/\gamma_2$